Personality Traits Predict Mental Health Across Three Universities: A Multi-Study Machine Learning Investigation (N = 1,559)

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Abstract

Big Five personality traits—particularly Neuroticism—are well-established correlates of internalizing psychopathology, yet multi-study replications using machine learning methods remain scarce, and the clinical utility of personality-based prediction has rarely been evaluated against passive behavioral sensing. We addressed these gaps across three independent university samples (Dartmouth, Notre Dame, University of Washington; N = 1,559) using a unified machine learning pipeline (Ridge, Elastic Net, Random Forest, SVR with Optuna tuning and SHAP interpretation) to predict depression, anxiety, stress, and loneliness from Big Five traits and passively sensed smartphone and wearable data. Neuroticism emerged as the top-ranked predictor in all 28 SHAP models spanning seven mental health instruments, with meta-analytic correlations of r = .63 for anxiety, r = .47 for stress, and r = .44 for depression. Personality-only models achieved AUC = .65–.80 for clinical-cutoff classification and R^2 = .09–.52 for continuous prediction, whereas behavior-only models performed near chance (AUC = .53–.60; $R^2 \leq 0$ in 12/15 comparisons). Passive sensing added significant incremental validity in only 1 of

8 tests after FDR correction ($\Delta R^2 = .024$). A notable exception was Study 2, where personality-plus-behavior models reached AUC = .83–.86, driven by communication metadata. SHAP rankings were virtually identical to OLS regression coefficients ($\tau = .94$), and multilayer perceptrons underperformed Ridge regression, confirming that the personality–mental health relationship is linear and that simpler models suffice. These findings position brief personality assessment—as few as two Neuroticism items—as a robust, low-cost mental health screening tool that outperforms weeks of continuous passive monitoring.

Keywords: Big Five personality, neuroticism, mental health prediction, passive sensing, SHAP, machine learning, replication

1 Introduction

1.1 Personality and Mental Health

The Five-Factor Model of personality (FFM; **??**) has emerged as one of the most robust frameworks for understanding individual differences in psychological functioning. Comprising five broad trait dimensions—Neuroticism, Extraversion, Openness, Agreeableness, and Conscientiousness—the FFM provides a comprehensive taxonomy of dispositional tendencies that has been replicated across cultures, age groups, and assessment methods over more than three decades of research. Among the most consequential applications of this framework has been the study of personality–psychopathology relations, where a large and consistent body of evidence demonstrates that personality traits are powerful predictors of mental health outcomes.

The landmark meta-analysis by **?**, synthesizing 175 studies and over 75,000 participants, established the empirical magnitude of these associations with considerable precision. Neuroticism emerged as the dominant predictor, associated with depressive disorders at $d = 1.65$ and with anxiety disorders at $d = 1.40$—effect sizes that rank among the largest in clinical psychology. The association is not limited to concurrent symptomatology. **?** argued that Neuroticism constitutes a transdiagnostic risk factor of substantial public health significance, prospectively predicting the onset of depression, anxiety, substance use disorders, and other forms of psychopathology across longitudinal studies spanning decades. Subsequent meta-analytic work confirmed that elevated Neuroticism approximately doubles the risk of developing a mental disorder even after adjusting for baseline symptoms and psychiatric history (**?**), supporting the interpretation that Neuroticism is not merely a correlate of distress but a genuine vulnerability factor.

Beyond Neuroticism, other FFM dimensions contribute meaningfully to the prediction of mental health. Conscientiousness has been identified as a protective factor, with meta-analytic evidence showing negative associations with depression ($d = -0.90$) and substance use disorders (**?**). Low Extraversion is associated with depression and so-

cial anxiety, while low Agreeableness shows links to externalizing psychopathology. **?** demonstrated that personality traits rival socioeconomic status and cognitive ability in predicting important life outcomes, including health and mortality. Theoretical accounts have conceptualized the personality–psychopathology relationship through both vulnerability models, in which traits predispose individuals to disorders, and spectrum models, in which traits and disorders represent different points on a shared continuum of functioning (**?**).

Despite this extensive literature, several important questions remain unresolved. First, the vast majority of personality–mental health research has relied on traditional statistical methods—zero-order correlations, hierarchical regression, structural equation modeling—rather than machine learning approaches that evaluate out-of-sample predictive accuracy. The distinction matters: a correlation of $r = .50$ in a training set does not guarantee that personality scores will generalize to predict symptoms in new individuals, particularly when multiple predictors are considered simultaneously and overfitting is a risk. Second, multi-study replications using identical analytic pipelines are rare. Most meta-analyses aggregate results across studies that differ in their measures, statistical methods, sample characteristics, and analytic choices, making it difficult to determine whether effect size heterogeneity reflects genuine population differences or merely methodological inconsistency. Third, clinical utility metrics—sensitivity, specificity, area under the ROC curve—are almost never reported, leaving unclear whether the statistical associations between personality and mental health translate into practically useful screening performance.

1.2 Passive Behavioral Sensing

While the personality–psychopathology literature has accumulated evidence over decades, a newer research tradition has sought to predict mental health using passively collected behavioral data from smartphones and wearable devices. **?** characterized the smartphone as "a potential game changer for psychiatry," envisioning continuous, passive collection of behavioral signals—movement patterns, sleep rhythms, social interactions, screen use—as a pathway to objective, real-time psychiatric assessment. This vision of *digital phenotyp-*

ing has been elaborated in comprehensive reviews (**???**) and has attracted substantial investment in dedicated research centers, government-funded initiatives, and commercial platforms.

Empirical support for the approach emerged from early studies such as **?**, who reported correlations between GPS-derived mobility features and PHQ-9 depression scores, and the landmark StudentLife project (**?**), which demonstrated that smartphone sensing could track behavioral changes across a college academic term. Subsequent work linked smartphone and wearable data to depression (**??**), anxiety (**?**), and general well-being (**?**). **?** outlined how smartphones could serve as "pocket laboratories" for psychological science, and **?** proposed that digital biomarkers could supplement or even supplant traditional symptom measures.

An implicit assumption underlying much of this work is that continuous behavioral data should provide information *beyond* what traditional questionnaires capture. Smartphones record what people *do*—their mobility, sleep, social interactions, digital habits—rather than what they report on subjective scales. The data are collected passively and continuously, requiring no ongoing participant effort. These features have made behavioral sensing an attractive complement—or potential replacement—for conventional self-report assessment. Yet the question of whether sensing data genuinely add predictive value over and above brief, well-validated questionnaires has received surprisingly little direct attention.

1.3 Gaps and Present Study

Despite the maturity of the personality–mental health literature and the rapid growth of digital phenotyping, several critical gaps remain at the intersection of these two traditions.

Gap 1: Multi-study replication with identical ML pipelines. Existing machine learning studies of personality and mental health are typically conducted in single samples, using study-specific analytic choices. The absence of multi-study replications with standardized pipelines makes it difficult to assess the robustness and generalizability of reported effects. Meta-analyses partially address this concern but cannot control for analytic heterogeneity

across primary studies.

Gap 2: Clinical utility evaluation. Although the statistical association between personality traits and mental health is well established, translation to clinical screening contexts—characterized by AUC, sensitivity, and specificity at established cutoffs—remains almost entirely absent from the literature. Without these metrics, it is unclear whether personality-based models can achieve performance levels useful for population-level mental health screening.

Gap 3: Incremental validity of behavioral sensing over personality. Studies in the digital phenotyping literature typically report sensing–outcome associations without including personality as a baseline predictor (**???**), while personality–psychopathology studies rarely incorporate passively sensed features (**??**). As a result, the incremental validity of behavioral sensing beyond personality—tested with appropriate multiple comparison correction—remains essentially unknown. Growing concerns about the robustness of sensing-based predictions reinforce this gap: **?** found that GPS-based depression models dropped from AUC = 0.82 to 0.57 in larger samples, and the GLOBEM benchmark (**?**) reported near-chance cross-dataset generalization. **?** attributed such failures to a fundamental *semantic gap* between low-level sensor signals and psychological constructs.

Gap 4: SHAP versus traditional methods. Interpretable machine learning techniques such as SHAP (**?**) have been applied to mental health prediction models, but it remains unclear whether data-driven feature importance rankings discover substantively different patterns than traditional regression coefficients. If SHAP rankings converge with standardized betas, this would suggest that the personality–mental health relationship is predominantly linear and that complex interpretability methods provide communicative rather than substantive gains.

The present study addresses these four gaps through a multi-study machine learning investigation of personality traits as predictors of mental health outcomes. We analyze three publicly available longitudinal datasets—StudentLife (**?**), NetHealth, and GLOBEM (**?**)—totaling $N = 1{,}559$ undergraduates across three U.S. universities (Dartmouth College, University of Notre Dame, University of Washington). All three datasets

contain both Big Five personality assessments and passive behavioral sensing data (smartphone and wearable), enabling direct comparison of these two measurement modalities within the same participants. A unified machine learning pipeline (Ridge regression, Elastic Net, Random Forest, SVR, and logistic regression with Optuna hyperparameter tuning, repeated cross-validation, and permutation testing) is applied identically across all datasets.

We organize our investigation around four aims:

Aim 1. Quantify the predictive accuracy of Big Five personality traits for mental health outcomes (depression, anxiety, stress, loneliness) using machine learning with out-of-sample evaluation, replicated across three independent studies.

Aim 2. Evaluate clinical utility by reporting AUC, sensitivity, and specificity at established clinical cutoffs for personality-based screening models.

Aim 3. Test the incremental validity of passive behavioral sensing beyond personality using nested model comparisons with false discovery rate correction.

Aim 4. Compare SHAP-derived feature importance rankings against traditional regression coefficients to assess whether machine learning interpretability reveals structure beyond what linear methods capture.

By applying an identical pipeline across three datasets, we provide a multi-study replication that moves beyond single-sample findings to assess the robustness and boundaries of personality-based mental health prediction. We further examine whether more complex models (multilayer perceptrons) improve upon simple penalized regression, whether demographic covariates attenuate personality effects, and whether the pattern of results holds across different mental health constructs and personality instruments (BFI-44 vs. BFI-10). The results speak to the practical question of what the simplest, most cost-effective approach to mental health screening might be—and whether the field's investment in continuous behavioral sensing provides meaningful returns beyond what a brief personality questionnaire already delivers.

2 Method

2.1 Datasets

We conducted secondary analyses of three publicly available longitudinal datasets from United States universities, comprising a total of $N = 1{,}559$ undergraduate participants. Table 1 summarizes the key characteristics of each study. All three studies received institutional ethical approval from their respective universities, and we refer readers to the original publications for detailed ethical review procedures.

2.1.1 Study 1: StudentLife (Dartmouth College)

The StudentLife dataset (**?**) comprises data from a 10-week smartphone sensing study conducted at Dartmouth College during the Fall 2013 academic term. Of 48 enrolled participants, $N = 28$ had complete Big Five personality and cumulative GPA records and formed our analytic sample. Participants were undergraduate and graduate students who carried Android smartphones instrumented with continuous sensing software throughout the term. Six outcome measures were available: cumulative GPA, depressive symptoms (PHQ-9), perceived stress (PSS), loneliness, flourishing, and negative affect (PANAS-NA).

2.1.2 Study 2: NetHealth (University of Notre Dame)

The NetHealth study is a large-scale longitudinal investigation of college student health conducted at the University of Notre Dame from 2015 to 2019. Participants were first-year students who wore Fitbit wearable devices and completed periodic surveys over multiple academic years. The full sample comprised $N = 722$ participants: 220 with both BFI-44 and GPA records, and 498 with both BFI-44 and at least one mental health instrument. Of these, $N = 365$ had complete behavioral sensing data (Fitbit activity, sleep, and communication logs). Demographic information was available for this study, including gender, native status, parental education, and family income. Mental health outcomes included the CES-D (depression), STAI-Trait (trait anxiety), and BAI (Beck

anxiety).

2.1.3 Study 3: GLOBEM (University of Washington)

The GLOBEM dataset (**?**) is a multi-year longitudinal study conducted at the University of Washington from 2018 to 2021, spanning four annual cohorts (INS-W_1 through INS-W_4). The total sample comprised $N = 809$ participants (800 with BFI-10 personality data; 786 with BDI-II depression scores). Participants wore Fitbit devices and carried smartphones instrumented with sensing software. The third cohort (INS-W_3, 2020–2021) overlapped with the COVID-19 pandemic, providing a natural test of robustness to environmental disruption. Each cohort included approximately 200 participants. Mental health outcomes included the BDI-II (depression), STAI-State (state anxiety), PSS-10 (perceived stress), CESD-10 (depression), and UCLA Loneliness Scale. Weekly PHQ-4 assessments were available for a subset ($N = 540$) over approximately 10 weeks, enabling trajectory analyses. No GPA data were available in this dataset.

2.2 Measures

2.2.1 Big Five Personality

Personality was assessed using the Big Five Inventory. Studies 1 and 2 used the 44-item BFI (BFI-44; **?**), which measures Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism on 5-point Likert scales. Internal consistency (Cronbach's α) in the present samples ranged from .67 to .86 (Study 1) and .69 to .87 (Study 2). Study 3 used the 10-item short form (BFI-10; **?**), which contains two items per dimension. Because the BFI-10 yields lower reliability ($\alpha \approx .65$), we conducted disattenuation analyses to assess the impact of measurement error on observed effect sizes (see Supplementary Materials).

2.2.2 Mental Health Outcomes

Multiple validated instruments assessed mental health across the three studies, providing convergent measurement of depression, anxiety, and stress constructs:

- **Depression.** The Patient Health Questionnaire-9 (PHQ-9; **?**) was used in Study 1 (9 items; clinical cutoff ≥ 10). The Center for Epidemiological Studies Depression Scale (CES-D; **?**) was used in Study 2 (20 items; cutoff ≥ 16). The Beck Depression Inventory–II (BDI-II; **?**) was used in Study 3 (21 items; mild cutoff ≥ 14, moderate cutoff ≥ 20). The 10-item CES-D (CESD-10) was also available in Study 3.

- **Anxiety.** The State-Trait Anxiety Inventory (STAI; **?**) was used in Studies 2 (Trait subscale; cutoff ≥ 45) and 3 (State subscale). The Beck Anxiety Inventory (BAI; **?**) was used in Study 2.

- **Stress.** The Perceived Stress Scale (PSS; **?**) was used in Study 1 (14 items). The PSS-10 (10-item version) was used in Study 3 (cutoff ≥ 20).

- **Other outcomes.** The UCLA Loneliness Scale (**?**) was available in Study 3. The PHQ-4 (**?**) provided weekly depression and anxiety screening in Study 3. Cumulative GPA from registrar records was available in Studies 1 and 2.

2.2.3 Behavioral Features

Passively sensed behavioral data were extracted from smartphones and wearable devices. The number and type of sensing modalities varied across studies, reflecting differences in the original study designs:

Study 1 collected data from 13 smartphone sensing modalities, including accelerometer-derived activity (stationary, walking, running), microphone-based audio classification (silence, voice, noise), conversation detection (frequency, duration), GPS location (entropy, time at home, distance traveled), screen usage (unlock frequency, duration), Bluetooth proximity (unique devices), Wi-Fi (unique access points), communication logs (calls, SMS), and app usage. This yielded 87 raw behavioral features, which were reduced to 8 orthogonal principal component analysis (PCA) composites capturing Mobility, Digital Usage, Audio Environment, Social Activity, Communication, Sleep, Screen Habits, and App Usage patterns.

Study 2 collected data from Fitbit wearables and communication logs, yielding 28

raw features: 11 activity features (daily steps, distance, calories, active minutes by intensity, sedentary minutes, step variability), 8 sleep features (total sleep time, time in bed, efficiency, awakenings, onset latency, variability, wake after sleep onset, regularity), 7 communication features (incoming/outgoing calls, call duration, SMS counts, unique contacts), and 2 quality indicators. These were reduced to 3 PCA composites: Activity-Sleep, Communication, and Quality.

Study 3 collected data from Fitbit wearables, smartphones, and GPS, yielding 19 main features: 4 activity features (daily steps, active minutes, sedentary minutes, step variability), 4 sleep features (total sleep time, efficiency, onset time, wake time), 4 call features (incoming/outgoing counts, duration, unique contacts), 3 screen features (on-frequency, duration, first/last unlock), and 4 location features (unique locations, entropy, time at home, distance traveled). These were reduced to 5 PCA composites: Activity, Sleep, Communication, Screen, and Location. As a sensitivity analysis, we also extracted 2,597 features using the RAPIDS pipeline, filtered to 1,258 after removing zero-variance columns.

PCA was applied within each study to address multicollinearity and the high feature-to-sample ratio. The number of retained components was determined by cumulative variance explained, and all PCA transformations were computed within cross-validation folds to prevent data leakage.

2.3 Analytic Strategy

All analyses were implemented in Python 3 using scikit-learn (**?**). The same analytic pipeline was applied uniformly across all three studies to facilitate direct comparison.

2.3.1 Machine Learning Pipeline

We evaluated four regression algorithms spanning linear and non-linear approaches: Ridge regression (**?**), Elastic Net (**?**), Random Forest (**?**), and Support Vector Regression (SVR; **?**). For each outcome, three feature sets were compared: personality traits only, behavioral composites only, and personality plus behavior combined.

Hyperparameters were tuned using Optuna Bayesian optimization (**?**) with a Tree-structured Parzen Estimator (TPE) sampler and 30 optimization trials. Cross-validation followed a repeated K-fold scheme: leave-one-out cross-validation (LOO-CV) for Study 1 ($N = 28$, where standard K-fold would yield unstable variance estimates) and 5-fold $\times$ 5-repeat cross-validation for Studies 2 and 3. Within each fold, features were standardized using `StandardScaler` fit exclusively on training data to prevent information leakage. For supplementary analyses, Ridge regression with a fixed regularization parameter ($\alpha = 1.0$) was used to ensure consistency across comparisons; for main analyses, `RidgeCV` with built-in cross-validated α selection was employed.

Model performance was assessed using out-of-fold R^2 (coefficient of determination). Statistical significance was evaluated via permutation tests (1,000 permutations for Studies 2 and 3; 199 permutations for Study 1 given computational constraints with LOO-CV), in which outcome labels were randomly shuffled and the full modeling pipeline re-executed to construct a null distribution of R^2 values.

2.3.2 SHAP Interpretability

To interpret model predictions, we computed Shapley Additive Explanations (SHAP; **?**) for every trained model. TreeSHAP (exact computation) was used for tree-based models (Random Forest), and KernelSHAP (model-agnostic approximation) was used for linear and kernel-based models (Ridge, Elastic Net, SVR). For each outcome, mean absolute SHAP values were computed across all test-fold predictions to rank feature importance. Cross-model consistency was quantified using Kendall's τ rank correlation between SHAP importance rankings across all model pairs. To validate SHAP against traditional methods, we compared SHAP-derived rankings with zero-order correlations and standardized OLS regression coefficients.

2.3.3 Clinical Classification

To evaluate clinical utility, we dichotomized mental health outcomes at established clinical cutoffs (CES-D $\geq$ 16, STAI $\geq$ 45, BDI-II $\geq$ 14 and $\geq$ 20, PSS $\geq$ 20) and trained

binary classifiers using Logistic Regression and Random Forest. Classification followed a repeated stratified 10-fold $\times$ 10-repeat cross-validation scheme to ensure stable estimates with preserved class proportions. Discrimination was assessed via area under the receiver operating characteristic curve (AUC) with 95% percentile bootstrap confidence intervals. Optimal classification thresholds were selected using Youden's J index (sensitivity + specificity -1), and sensitivity at fixed specificity of 0.80 was reported for clinical relevance.

2.3.4 Incremental Validity

To test whether behavioral sensing adds predictive value beyond personality, we conducted nested F-tests comparing personality-only models against personality-plus-behavior models within an OLS regression framework. The null hypothesis was that the additional behavioral predictors do not explain variance beyond personality traits. Because multiple comparisons were conducted across outcomes and studies, p-values were corrected using the Benjamini–Hochberg false discovery rate (BH-FDR; **?**) procedure across all tests.

2.3.5 Meta-Analysis

To synthesize effect sizes across studies, we conducted random-effects meta-analyses using the DerSimonian–Laird estimator (**?**). Pearson correlations between personality traits and outcomes were transformed to Fisher's z prior to pooling, and pooled estimates were back-transformed for interpretation. Heterogeneity was assessed using the I^2 statistic, where values of 25%, 50%, and 75% indicate low, moderate, and high heterogeneity, respectively (**?**). Forest plots display study-level and pooled effect sizes with 95% confidence intervals.

2.3.6 Demographic Controls

In Study 2, where demographic data were available, we conducted hierarchical regression analyses to assess whether personality effects survive after controlling for demographic confounders. The regression was structured in three steps: (1) demographics only (gen-

der, native status, parental education, family income), (2) demographics plus personality traits, and (3) demographics plus personality plus behavioral composites. The incremental ΔR^2 at each step was tested for significance. Study 3 did not collect demographic data beyond cohort membership, and Study 1 was too small ($N = 28$) to support covariate-adjusted models.

3 Results

3.1 Head-to-Head: Personality Questionnaires vs Passive Sensing

Table 2 presents the primary comparison: out-of-fold R^2 with 95% bootstrap confidence intervals for personality-only versus sensing-only models across all 15 study–outcome combinations in three datasets. Personality questionnaires outperformed passive behavioral sensing in 14 of 15 comparisons (93%). The median advantage was $\Delta R^2 = .21$, with personality-only R^2 values ranging from near zero (S2 Loneliness: $R^2 = .01$) to substantial (S2 STAI-Trait: $R^2 = .52$ [95% CI: .41, .62]). In contrast, sensing-only models produced $R^2 \leq 0$ in 12 of 15 comparisons, indicating worse-than-chance prediction.

The strongest personality effects emerged for anxiety-related constructs. In Study 2, personality explained 52.2% of variance in trait anxiety ($R^2 = .52$ [.41, .62]) and 27.9% in depression ($R^2 = .28$ [.09, .39]). In Study 3, personality explained 20.0% of variance in state anxiety ($R^2 = .20$ [.08, .26]) and 14.3% in perceived stress ($R^2 = .14$ [−.00, .22]). Even in the smallest study ($N = 27$), personality predicted perceived stress ($R^2 = .39$ [.01, .72]) and loneliness ($R^2 = .20$ [−.19, .50]).

Behavioral sensing produced only one positive R^2 value across all studies: S1 PHQ-9 ($R^2 = .20$ [−.65, .82]), the sole comparison in which sensing outperformed personality. This was also the only outcome for which personality yielded a negative R^2 (−.55), likely attributable to overfitting in the small sample ($N = 27$) with leave-one-out cross-validation. Notably, the confidence intervals for both estimates in this comparison were extremely wide, spanning from negative to positive values, underscoring the instability

of estimates at this sample size.

For academic performance, personality predicted GPA in Study 2 ($R^2 = .002$ [$-.32$, .10]) and Study 1 ($R^2 = -.17$), though neither estimate was statistically distinguishable from zero. Sensing performed even worse for GPA, with $R^2 = -1.85$ in Study 1 and $R^2 = -.04$ in Study 2.

3.2 Meta-Analytic Synthesis

Table 3 and Figure 1 present random-effects meta-analytic estimates pooling personality–outcome correlations across studies. Sensing features were not meta-analyzed because $R^2 \leq 0$ in most study–outcome combinations, precluding meaningful pooled estimates.

The strongest meta-analytic effect was for Neuroticism predicting anxiety and stress (pooled $r = .63$ [95% CI: .38, .80], $k = 3$, $N = 1{,}324$, $p < .001$), followed by Neuroticism predicting depression (pooled $r = .44$ [.24, .61], $k = 3$, $N = 1{,}304$, $p < .001$). Conscientiousness showed a moderate negative association with depression (pooled $r = -.21$ [$-.29$, $-.12$], $k = 3$, $N = 1{,}304$, $p < .001$) with comparatively low heterogeneity ($I^2 = 44.8\%$, $p = .163$). Conscientiousness also predicted GPA (pooled $r = .38$ [.06, .62], $k = 2$, $N = 248$, $p = .019$), demonstrating trait-specific dissociation: Neuroticism was the primary predictor of mental health, whereas Conscientiousness was the primary predictor of academic performance.

Heterogeneity was high for most Neuroticism associations: $I^2 = 91.8\%$ for Neuroticism–Depression ($p < .001$), $I^2 = 96.2\%$ for Neuroticism–Anxiety ($p < .001$), and $I^2 = 94.4\%$ for Neuroticism–Loneliness ($p < .001$). The Neuroticism–Loneliness association was not significant (pooled $r = .13$ [$-.15$, .40], $p = .348$), reflecting inconsistent directions across studies (S1: $r = .50$; S2: $r = -.14$; S3: $r = .17$). Extraversion–Loneliness was similarly non-significant and heterogeneous (pooled $r = -.14$ [$-.46$, .21], $I^2 = 96.6\%$). The Neuroticism–Perceived Stress association was large but imprecise (pooled $r = .58$ [.07, .85], $k = 2$, $I^2 = 88.1\%$), reflecting the combination of a very strong effect in Study 1 ($r = .76$, $N = 27$) with a more moderate effect in Study 3 ($r = .37$, $N = 800$).

The high heterogeneity in most associations likely reflects genuine differences across

studies in sample characteristics, personality measure length (BFI-44 vs. BFI-10), and outcome instruments, rather than sampling error alone. Despite this heterogeneity, the direction of effects was consistent for Neuroticism–Depression and Neuroticism–Anxiety across all three studies.

3.3 Clinical Binary Classification

Table 4 presents classification performance for detecting clinically elevated symptom levels. We evaluated personality-only, personality-plus-behavior, and behavior-only models using logistic regression and random forest classifiers with repeated stratified cross-validation.

In Study 2, where both personality and behavioral data were available for the same participants, personality-only models achieved AUC = .75 [95% CI: .62, .88] for CES-D $\geq$ 16 and AUC = .79 [.68, .90] for STAI $\geq$ 45. Adding behavioral features improved discrimination modestly: AUC = .83 [.70, .93] for CES-D $\geq$ 16 and AUC = .86 [.71, .96] for STAI $\geq$ 45. However, behavior-only models performed near chance (AUC = .55 [.40, .70] and AUC = .58 [.42, .75]), confirming that personality carried the predictive signal.

In Study 3, classification performance was more modest. Personality-only models achieved AUC = .65 [.55, .76] for BDI-II $\geq$ 14, AUC = .66 [.55, .79] for BDI-II $\geq$ 20, and AUC = .69 [.58, .77] for PSS $\geq$ 20. Combined personality-plus-behavior models showed marginal improvement (AUC = .67 [.52, .78] for BDI-II $\geq$ 14, AUC = .69 [.52, .82] for BDI-II $\geq$ 20, AUC = .70 [.59, .81] for PSS $\geq$ 20). Behavior-only models again performed near chance (AUC = .57–.60).

DeLong tests comparing personality-only versus personality-plus-behavior AUC were conducted for all five classification targets. None survived Benjamini–Hochberg FDR correction (0/5; all $p_{\text{FDR}} > .44$), indicating that the observed improvements from adding behavioral sensing were not statistically reliable.

Sensitivity and specificity patterns were consistent across studies. At optimal Youden thresholds, personality-only models achieved high sensitivity (range: .71–.85) at the cost of moderate specificity (range: .60–.70). In Study 2, the combined model for STAI $\geq$

45 showed the best balance: sensitivity = .85 [.56, 1.00] and specificity = .80 [.57, 1.00], with PPV = .63 and NPV = .95, suggesting potential utility as a screening instrument in this specific context.

3.4 Incremental Validity of Behavioral Sensing

Table 5 reports nested F-tests evaluating whether behavioral sensing composites add significant variance beyond personality traits in OLS regression. Across eight study–outcome combinations in Studies 2 and 3 (Study 1 was excluded due to insufficient sample size), only one survived Benjamini–Hochberg FDR correction: S3 BDI-II ($\Delta R^2 = .024$, $F(5, 577) = 3.31$, $p_{\text{FDR}} = .047$). The remaining seven tests showed non-significant increments, with ΔR^2 values ranging from .001 (S2 BAI) to .018 (S3 STAI-State and S3 CESD), all $p_{\text{FDR}} > .08$.

Even in the one significant case, the incremental variance was small in absolute terms ($\Delta R^2 = .024$), representing a 2.4 percentage-point increase from personality-only $R^2 = .130$ to full-model $R^2 = .154$. The overall pattern was clear: behavioral sensing added negligible predictive information beyond what five self-report personality items already captured.

In Study 2, behavioral sensing contributed $\Delta R^2 = .009$ for CES-D, $\Delta R^2 = .002$ for STAI-Trait, and $\Delta R^2 = .003$ for BAI, none approaching significance after FDR correction. This was notable given that Study 2 had the richest behavioral feature set (Fitbit activity, sleep, and communication data) and the largest combined sample ($N = 363$ with complete behavioral data).

3.5 Methodological Checks

Several supplementary analyses assessed the robustness of the primary findings (see Supplementary Materials for full results of all 41 analyses).

3.5.1 SHAP and OLS Agreement

SHAP-derived feature importance rankings closely matched standardized OLS regression coefficients. Across seven personality–outcome combinations in Studies 2 and 3, Kendall's τ between β and SHAP rankings ranged from .80 to 1.00 (mean $\tau = .94$). Top-ranked feature agreement was perfect: Neuroticism was identified as the strongest predictor by both methods in 6 of 7 comparisons, with Extraversion ranked first for UCLA Loneliness. This convergence indicates that the personality–mental health relationship is predominantly linear, and that SHAP values provide communicative rather than substantively novel information over traditional regression weights.

3.5.2 Neural Network Comparison

A two-layer multilayer perceptron (MLP) with Optuna-tuned hyperparameters consistently underperformed Ridge regression across all outcomes (Table S1). In Study 1, MLP showed catastrophic overfitting ($R^2 = -4.06$ for Flourishing; $R^2 = -.47$ for Loneliness). In Studies 2 and 3, MLP R^2 values were systematically lower than Ridge: for example, S2 STAI-Trait MLP $R^2 = .49$ versus Ridge $R^2 = .52$; S3 STAI-State MLP $R^2 = .14$ versus Ridge $R^2 = .20$. This pattern reinforces the conclusion that the personality–mental health association is linear, and that additional model complexity introduces noise without capturing meaningful non-linear structure.

3.5.3 Demographic Controls

In Study 2, where demographic covariates (gender, native status, parental education, family income) were available, personality traits continued to explain substantial incremental variance beyond demographics: $\Delta R^2 = .33$ for CES-D, $\Delta R^2 = .52$ for STAI-Trait, and $\Delta R^2 = .20$ for BAI (all $p_{\text{FDR}} < .001$). In contrast, behavioral sensing added $\Delta R^2 < .01$ beyond demographics and personality for all three outcomes (all $p_{\text{FDR}} > .24$). These results confirm that personality effects are not reducible to demographic confounds and that sensing provides no incremental benefit even after accounting for shared demographic variance.

3.5.4 Cross-Study Consistency

Conscientiousness was identified as the top SHAP predictor of GPA across all four model types (Elastic Net, Ridge, Random Forest, SVR) in both Studies 1 and 2, demonstrating trait-specific consistency: Neuroticism dominated mental health prediction while Conscientiousness dominated academic prediction. Furthermore, Neuroticism was ranked as the top predictor for all mental health outcomes across all model types in Studies 2 and 3 (8/8 study–construct–model combinations), confirming the robustness of this finding to analytic choices.

4 Discussion

Across three independent university samples totaling N = 1,559 participants, Big Five personality traits—and Neuroticism in particular—emerged as robust, replicable predictors of mental health outcomes. These findings converge across seven self-report instruments, four machine learning algorithms, and universities spanning different regions and time periods. We discuss the implications in five sections, addressing the centrality of Neuroticism, the limited added value of passive behavioral sensing, the sufficiency of simple models, the boundaries of our evidence, and directions for future research.

4.1 Neuroticism as a Transdiagnostic Marker

The most consistent finding across all analyses was the dominance of Neuroticism as a predictor of mental health. Neuroticism ranked as the most important feature in all 28 SHAP models estimated across the three studies, spanning depression (PHQ-9, CES-D, BDI-II), anxiety (STAI, BAI), stress (PSS-10), and loneliness (UCLA Loneliness Scale). Meta-analytic aggregation confirmed large pooled effects: $r = .63$ for anxiety, $r = .44$ for depression, and $r = .47$ for stress. These effect sizes align with **?**, whose meta-analysis of 175 studies identified Neuroticism as the strongest personality correlate of internalizing disorders ($d = 1.65$ for depression, $d = 1.01$ for anxiety). Our contribution is not the direction of this association—which has been well-established since at least **?**—but its

consistency across a machine learning framework that makes no parametric assumptions about the form of the personality–outcome relationship.

The trait specificity of the findings strengthens the case for Neuroticism as a trans-diagnostic vulnerability marker (**?**). Conscientiousness emerged as the dominant predictor of academic performance (GPA) in all eight models where GPA was an outcome, whereas Neuroticism was the dominant predictor of every mental health outcome—a double dissociation that was never reversed across studies. This pattern is consistent with a theoretical framework in which Conscientiousness captures behavioral regulation relevant to academic effort and persistence, while Neuroticism captures emotional reactivity and negative affectivity that cut across diagnostic categories (**?**). That two items from the BFI—"Is depressed, blue" and "Is emotionally stable, not easily upset"—explained $R^2 = .36$ for CES-D depression scores underscores the clinical efficiency of Neuroticism: a 10-second self-report captured more between-person variance than weeks of continuous passive sensing ($R^2 = -.16$ for 28 behavioral features).

The robustness of these findings across measurement contexts is noteworthy. The Neuroticism–mental health association held whether personality was measured with the 44-item BFI (**?**) (Studies 1 and 2) or the abbreviated BFI-10 (**?**) (Study 3), despite the latter's reduced reliability ($\alpha \approx .65$). Controlling for gender, race, native-English-speaker status, parental education, and family income reduced personality's predictive contribution only modestly (personality $\Delta R^2 = .20$–$.52$ after demographic controls, versus demographics alone $R^2 = .03$–$.06$). Excluding the pandemic-era cohort (2020–2021) produced negligible changes ($\Delta R^2 < .015$), confirming that the association is not an artifact of COVID-related distress.

4.2 The Limited Value of Passive Behavioral Sensing

Despite substantial investment in digital phenotyping as a mental health monitoring tool (**??**), passive behavioral sensing contributed minimally to prediction in our data. Behavior-only models produced $R^2 \leq 0$ in 12 of 15 outcome comparisons—performing worse than predicting the sample mean. In clinical classification, behavior-only AUC

ranged from .53 to .60, barely above chance, compared to personality-only AUC of .65 to .79. Formal incremental validity testing revealed that only 1 of 8 comparisons survived Benjamini–Hochberg FDR correction (Study 3 BDI-II, $\Delta R^2 = .024$, $p_{\text{FDR}} = .047$)—a statistically significant but clinically negligible contribution.

The exception merits discussion. In Study 2, the combined personality-plus-behavior model achieved AUC = .83–.86 for depression and trait anxiety, a meaningful improvement over personality alone (AUC = .75–.79). This gain was driven by communication metadata—call frequency and SMS patterns from the NetHealth dataset—features that capture social interaction, which is more semantically proximal to mental health than step counts or sleep duration. Study 3, which lacked communication features, showed no comparable benefit. This pattern suggests that when sensing captures social behavior rather than mere physical activity, it can provide marginal incremental information, but the conditions for this contribution are restrictive.

We emphasize that this is not a technology failure per se. Rather, it reflects what **?** termed a "semantic gap" between low-level sensor signals and the higher-order psychological constructs they are intended to index. Personality questionnaires directly probe stable dispositional tendencies that are theoretically linked to psychopathology; passive sensing captures moment-to-moment behavioral fluctuations influenced by weather, academic schedules, and social obligations in addition to any underlying trait signal. Our dose-response analysis confirmed that extending data collection from 7 to 92 days did not improve sensing performance, suggesting the between-person signal is not buried in short measurement windows but is fundamentally absent at the population level. Converging evidence from **?**, who found GPS-based depression prediction dropped from AUC = .82 to .57 in large samples, and from the GLOBEM benchmark (**?**), which reported near-chance cross-cohort generalization, supports this interpretation.

4.3 Simpler Models Suffice

A methodologically important finding is that analytic complexity provided no benefit for the personality–mental health prediction task. SHAP-derived feature importance rank-

ings were virtually identical to standardized OLS regression coefficients (mean Kendall's $\tau = .94$ across seven personality–outcome comparisons), indicating that penalized linear models captured the same structure that ordinary regression reveals. Multilayer perceptrons consistently underperformed Ridge regression, often catastrophically so in smaller samples (Study 1 MLP $R^2 = -4.06$ for Flourishing). Optuna hyperparameter optimization, applied systematically across all model types, did not uncover configurations that materially improved performance.

These results suggest that the personality–mental health relationship is predominantly linear: five broad trait dimensions predict internalizing symptoms through additive main effects, with no meaningful interactions or non-linear thresholds for machine learning to exploit. This finding has practical implications. SHAP analysis, while computationally expensive, served primarily a communication function—producing intuitive visualizations of feature importance—rather than revealing structure invisible to traditional regression. For personality-based mental health prediction, Ridge or Elastic Net regression with cross-validated regularization appears sufficient. The field's investment in increasingly complex architectures (deep learning, transformers, attention mechanisms) may be misaligned with the low-dimensional, linear nature of the personality–outcome signal (**?**).

4.4 Limitations

Several limitations constrain the generalizability of our findings. First, all three samples comprise undergraduate students at selective U.S. universities (Dartmouth, Notre Dame, University of Washington), restricting inference to relatively affluent, young, and predominantly White populations. Whether personality's predictive dominance extends to clinical populations, older adults, or non-Western cultural contexts—where the structure of personality and its relation to psychopathology may differ—remains an open question.

Second, our design is cross-sectional with respect to the core prediction models. Although supplementary longitudinal and trajectory analyses confirmed that personality predicts mean symptom levels but not within-person change over time—consistent with a trait interpretation—we cannot make causal claims about the personality–mental health

association. Prospective designs that assess personality before disorder onset would provide stronger evidence for the screening utility we propose.

Third, Study 3 used the BFI-10 (**?**), which has lower internal consistency ($\alpha \approx .65$) than the BFI-44 used in Studies 1 and 2. Attenuation due to measurement error biases results *against* personality, and our disattenuation analyses confirmed that reliability-corrected effects are larger than observed. Nevertheless, the use of an abbreviated instrument in the largest study introduces measurement imprecision.

Fourth, we analyzed secondary data not originally designed for this specific investigation. Sensing modalities differ across studies—Study 2 included Fitbit wearable and communication metadata, while Studies 1 and 3 relied primarily on smartphone sensing—and no single study captures all possible behavioral domains (e.g., GPS, social media, physiological signals). The absence of these modalities means we cannot rule out the possibility that unmeasured behavioral features carry stronger predictive signal.

Fifth, all mental health outcomes are self-report symptom scales, not clinical diagnoses. Questionnaire cutoffs (e.g., CES-D $\geq$ 16, BDI-II $\geq$ 14) approximate but do not equate to DSM-5 diagnostic criteria, and predictive performance in clinical diagnostic contexts may differ.

Finally, Study 1's small sample (N = 27–28) was treated as a discovery dataset throughout. Its results are informative for hypothesis generation but should not be interpreted as confirmatory evidence.

4.5 Implications and Future Directions

Our findings carry implications for both clinical practice and research methodology. First, personality screening represents a low-cost, scalable triage tool for identifying students at elevated risk for mental health difficulties. A two-item Neuroticism assessment, requiring approximately 10 seconds to complete, explained more variance in depression than weeks of continuous behavioral monitoring. In resource-constrained settings—university counseling centers, primary care waiting rooms, large-scale online surveys—personality screening could serve as a first-pass risk stratification tool, directing clinical resources

toward individuals with the highest trait vulnerability (**?**).

Second, we recommend that digital phenotyping studies routinely include a personality questionnaire baseline. Without this comparator, it is impossible to determine whether sensing provides unique information or merely recapitulates variance already accessible through a brief self-report. Our meta-analytic personality benchmarks (R^2 = .09–.53 across outcomes) provide reference values against which sensing contributions can be evaluated.

Third, future research should prioritize longitudinal prediction: does personality assessed at university entrance predict symptom trajectories across semesters, and can sensing add incremental value for detecting within-person deviations from trait-predicted baselines? Our idiographic analyses suggest that 17% of individuals show meaningful person-specific sensing–mood associations, hinting that sensing's value may lie in personalized monitoring of high-risk individuals identified by initial trait assessment, rather than population-level screening.

5 Conclusion

Across three universities and 1,559 participants, Big Five personality traits—led by Neuroticism—robustly predicted depression, anxiety, stress, and loneliness, replicating across seven instruments, four machine learning algorithms, and diverse demographic contexts. Neuroticism emerged as the top-ranked predictor in all 28 SHAP models, with meta-analytic correlations of r = .44–.63, while passive behavioral sensing added negligible incremental validity (1/8 comparisons significant after FDR correction). The personality–mental health relationship was predominantly linear, with Ridge regression matching or exceeding more complex models including multilayer perceptrons. These findings position brief personality assessment as a robust, low-cost mental health screening tool and suggest that the field's pursuit of complex sensing and modeling architectures may be less productive than leveraging the well-established, efficient, and replicable signal carried by a small number of personality trait items.

Data Availability

Study 1 (StudentLife) data are available from the StudentLife project (`https://studentlife.cs.dartmouth.edu/`). Study 2 (NetHealth) data are available via `https://nethealth.nd.edu/` with registration. Study 3 (GLOBEM) data are available via PhysioNet (`https://physionet.org/content/globem/`) with credentialed access and a data use agreement. Analysis code is available at [GitHub URL].

Ethics Statement

All three studies received institutional ethical approval from their respective universities (Dartmouth College, University of Notre Dame, University of Washington). Informed consent was obtained from all participants in the original studies. This secondary analysis was conducted under [NUS IRB reference].

Author Contributions (CRediT)

[Author A]: Conceptualization, Data curation, Formal analysis, Methodology, Software, Visualization, Writing – original draft. **Cyrus S. H. Ho:** Supervision, Writing – review & editing.

Conflicts of Interest

The authors declare no competing interests.

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Table 1: Overview of the Three Study Datasets

	Study 1	**Study 2**	**Study 3**
Source	StudentLife	NetHealth	GLOBEM
University	Dartmouth	Notre Dame	U. Washington
Year	2013	2015–2019	2018–2021
N	28	722	809
BFI measure	BFI-44	BFI-44	BFI-10
Sensing	Smartphone (13 modalities)	Fitbit + Communication	Smartphone (RAPIDS)
Duration	10 weeks	4 years	1 year (4 cohorts)
Depression	PHQ-9	CES-D	BDI-II, PHQ-4, PHQ-9
Anxiety	PSS	STAI, BAI	GAD-7, PHQ-4
GPA	Yes	Yes	No

Note. BFI = Big Five Inventory. RAPIDS = Reproducible Analysis Pipeline for Data Streams. PSS = Perceived Stress Scale. STAI = State-Trait Anxiety Inventory. BAI = Beck Anxiety Inventory. CES-D = Center for Epidemiologic Studies Depression Scale. BDI-II = Beck Depression Inventory-II. PHQ = Patient Health Questionnaire. GAD-7 = Generalized Anxiety Disorder 7-item scale.

Table 2: Grand Synthesis: Personality vs. Sensing R^2 with 95% Confidence Intervals

Study	**Outcome**	**N**	**Personality R^2 [95% CI]**	**Sensing R^2 [95% CI]**	**Winner**	ΔR^2
S1	PHQ-9	27	−.55 [−2.34, .24]	.20 [−.65, .82]	Sensing	.75
S1	PSS	27	.39 **[.01, .72]**	−.23 [−1.12, .22]	**Pers.**	.62
S1	Loneliness	27	.20 [−.19, .50]	−.25 [−.69, .25]	Pers.	.46
S1	GPA	27	−.17 [−.84, .26]	−1.85 [−5.58, −.28]	Pers.	1.68
S2	CES-D	498	.279 **[.088, .391]**	−.011 [−.085, .051]	**Pers.**	.290
S2	STAI-Trait	498	.522 **[.409, .620]**	.002 [−.090, .074]	**Pers.**	.520
S2	BAI	498	.176 **[−.025, .334]**	−.035 [−.127, .017]	**Pers.**	.211
S2	Loneliness	498	.011 [−.103, .063]	−.007 [−.089, .055]	Pers.	.018
S2	Self-Esteem	717	.412 **[.260, .495]**	−.008 [−.062, .046]	**Pers.**	.420
S2	GPA	220	.002 [−.320, .103]	−.044 [−.274, .050]	Pers.	.046
S3	BDI-II	779	.091 **[−.008, .167]**	−.001 [−.089, .064]	**Pers.**	.092
S3	STAI-State	799	.200 **[.078, .260]**	−.007 [−.093, .043]	**Pers.**	.208
S3	PSS-10	800	.143 **[−.002, .221]**	−.009 [−.087, .032]	**Pers.**	.152
S3	CESD	800	.093 **[−.000, .167]**	−.010 [−.083, .032]	**Pers.**	.103
S3	UCLA	798	.092 **[.005, .149]**	−.023 [−.082, .016]	**Pers.**	.115

Note. R^2 values estimated via cross-validated ridge regression; negative R^2 indicates prediction worse than intercept-only model. 95% CIs obtained by bootstrap (B = 1,000). Bold indicates personality R^2 CI excludes zero. Pers. = Personality; ΔR^2 = Personality R^2 − Sensing R^2. Personality wins in 14 of 15 comparisons.

Table 3: Random-Effects Meta-Analysis: Neuroticism–Mental Health Correlations Across Studies

Trait–Outcome	k	N	**Pooled r [95% CI]**	I^2 (%)	p
N → Depression	3	1,304	.444 **[.235, .614]**	91.8	< .001
N → Anxiety/Stress	3	1,324	.632 **[.384, .795]**	96.2	< .001
N → Loneliness	3	1,323	.135 [−.147, .396]	94.4	.348
E → Loneliness	3	1,323	−.144 [−.465, .209]	96.6	.425
C → Depression	3	1,304	−.209 **[−.291, −.124]**	44.8	< .001
N → Perceived Stress	2	827	.576 **[.071, .846]**	88.1	.028
C → GPA	2	248	.376 **[.064, .620]**	64.0	.019

Note. k = number of studies; I^2 = heterogeneity index (proportion of variance due to between-study differences); N = Neuroticism; E = Extraversion; C = Conscientiousness. Bold indicates $p < .05$. High I^2 values reflect genuine between-site variation rather than methodological inconsistency. Random-effects model (DerSimonian–Laird).

Table 4: Clinical Classification Performance: Personality vs. Sensing Feature Sets

Study / Outcome	Features	AUC [95% CI]	Sensitivity [95% CI]	Specificity [95% CI]
Study 2 — CES-D $\geq$ 16				
	Pers-only	.751 **[.622, .882]**	.849 [.500, 1.00]	.634 [.364, .861]
	Pers+Beh	.833 **[.698, .932]**	.839 [.495, 1.00]	.770 [.489, 1.00]
	Beh-only	.552 [.402, .704]	.648 [.139, 1.00]	.607 [.112, .982]
Study 2 — STAI-Trait $\geq$ 45				
	Pers-only	.795 **[.676, .896]**	.855 [.615, 1.00]	.705 [.487, .898]
	Pers+Beh	.856 **[.707, .960]**	.852 [.556, 1.00]	.798 [.571, 1.00]
	Beh-only	.577 [.416, .749]	.702 [.200, 1.00]	.592 [.160, .963]
Study 3 — BDI-II $\geq$ 14				
	Pers-only	.654 **[.552, .763]**	.708 [.323, .985]	.604 [.265, .919]
	Pers+Beh	.671 **[.520, .778]**	.732 [.373, .938]	.632 [.374, .890]
	Beh-only	.569 [.446, .658]	.680 [.228, 1.00]	.542 [.124, .917]
Study 3 — BDI-II $\geq$ 20				
	Pers-only	.667 **[.548, .788]**	.718 [.368, 1.00]	.623 [.361, .935]
	Pers+Beh	.686 **[.515, .825]**	.710 [.259, .929]	.682 [.340, .957]
	Beh-only	.604 [.458, .722]	.650 [.308, 1.00]	.637 [.182, .957]
Study 3 — PSS $\geq$ 20				
	Pers-only	.686 **[.575, .773]**	.721 [.404, .937]	.633 [.366, .914]
	Pers+Beh	.704 **[.594, .810]**	.680 [.433, .935]	.718 [.383, .931]
	Beh-only	.605 [.484, .712]	.664 [.233, .966]	.590 [.262, .933]

Note. AUC = area under the receiver operating characteristic curve. Pers-only = Big Five personality traits only; Beh-only = passive sensing behavioral features only; Pers+Beh = combined model. CIs obtained by bootstrap (B = 1,000). Best model selected by cross-validated AUC (LR = logistic regression, RF = random forest). Bold indicates AUC > .65.

Table 5: Incremental Validity: Does Sensing Add Predictive Value Beyond Personality?

Study	Outcome	N	R^2_{pers}	R^2_{full}	ΔR^2	F	p	Sig.[a]
S2	CES-D	363	.357	.367	.009	1.75	.157	
S2	STAI-Trait	363	.568	.570	.002	0.68	.566	
S2	BAI	363	.245	.248	.003	0.43	.734	
S3	BDI-II	588	.130	.154	.024	3.31	.006	*
S3	STAI-State	593	.226	.244	.018	2.71	.020	
S3	PSS-10	594	.176	.190	.014	1.99	.078	
S3	CESD	594	.112	.130	.018	2.39	.036	
S3	UCLA Loneliness	592	.108	.116	.008	1.08	.370	

Note. $R^2_{\text{pers}} = R^2$ from personality-only model; $R^2_{\text{full}} = R^2$ from personality + sensing model; ΔR^2 = incremental variance explained by sensing features. F-test evaluates whether sensing features add significant predictive value over personality alone. [a]Significance after FDR correction ($q < .05$): * $p_{\text{FDR}} < .05$. S2 has 3 behavioral features; S3 has 5 behavioral features.